

Space-Based Cybersecurity

Securing the Orbital Edge and Inter-Satellite Links

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1. Securing the "Orbital Edge"

In 2026, the network perimeter has officially left the atmosphere. With the massive expansion of Low Earth Orbit (LEO) constellations, we are no longer just securing terrestrial data centers; we are securing the Orbital Edge. These satellites are no longer "bent pipes" that simply reflect signals; they are high-compute nodes running containerized microservices and AI-driven routing agents.

The challenge is that a satellite is physically inaccessible once launched. If a vulnerability is discovered in the flight software, "patching" requires a high-stakes remote update over a high-latency, limited-bandwidth link. Furthermore, the physical security of the "ground station" remains the weakest link, as compromising a single uplink can potentially provide a foothold into an entire constellation.

2. The Threat Actor Playbook: "Ground-to-Orbit Hijacking"

Adversaries are moving beyond simple signal jamming to more sophisticated logical attacks targeting the telemetry, tracking, and command (TT&C) channels.

The Tactic: Ephemeris Spoofing and Shadow Routing

1. Ephemeris Manipulation: Attackers inject false orbital data (Ephemeris) into the ground control system. This causes the satellite to miscalculate its position, leading to "pointing errors" that drop legitimate connections.
2. Shadow Routing: By exploiting vulnerabilities in the inter-satellite routing protocols, an attacker can redirect traffic through a compromised satellite node.
3. The "Silent" Exfiltration: Unlike a terrestrial breach, shadow routing in space is incredibly difficult to detect. The data is intercepted in the vacuum of space, far from the reach of traditional deep-packet inspection (DPI) tools located at terrestrial chokepoints.

3. Inter-Satellite Link (ISL) and PQC

The most critical advancement in 2026 is the use of Optical Inter-Satellite Links (OISL), lasers, to pass data between nodes. While lasers are inherently harder to intercept than traditional RF, they are not immune to sophisticated "light-tapping" or "side-channel" analysis.

Hybrid PQC in Vacuum

To defend against "Harvest Now, Decrypt Later" (HNDL) threats in the space segment, we are implementing Hybrid Post-Quantum Cryptography (PQC) for all ISL handshakes. This combines classical ECDH with lattice-based algorithms like ML-KEM.

The signal-to-noise ratio (SNR) required for a successful quantum-safe handshake in an optical link is defined by the Link Budget:

$$SNR = \frac{P_r}{N_0 \cdot B}$$

Where:

- P_r is the received optical power.
- N_0 is the noise spectral density (primarily solar background in space).
- B is the bandwidth of the signal.

In 2026, we use Ephemeris-Based Authentication. The satellite will only accept a handshake if the incoming signal's Doppler shift (Δf) matches the predicted relative velocity (v_{rel}) of the expected neighbor:

$$\Delta f = f_0 \left(\frac{v_{rel}}{c} \right)$$

4. The "Above-the-Cloud" Security Model

We used to talk about "Cloud Security," but we are now entering the era of Above-the-Cloud Security. The traditional Zero-Trust model assumes that the "wire" is compromised. In space, we must assume that the entire orbital plane is a hostile environment.

Every satellite must be its own Silicon Root of Trust (SRoT). If a satellite cannot verify its own internal state via TPM 3.0 during its "Measured Boot" sequence, it must autonomously decommission itself before it can be used as a pivot point for a wider constellation breach. The "High Ground" is only an advantage if you can keep it secure.